

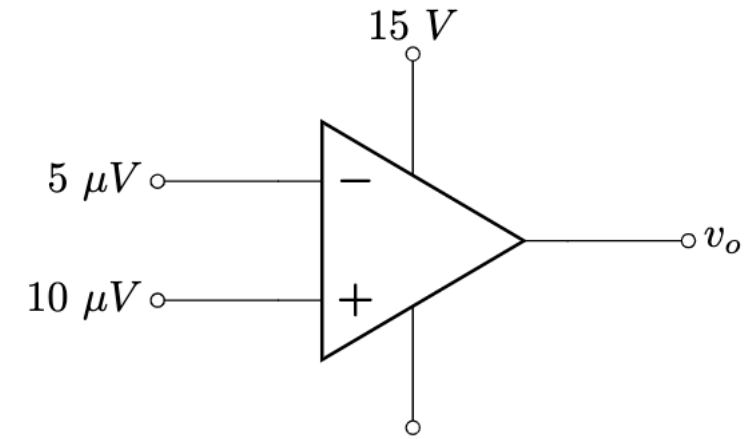
# Outline

- Feedback in Op-Amp circuit
- Negative Feedback
- Open Loop VS Closed Loop Gain
- Closed Loop Configuration

# Types of Op-Amp configuration

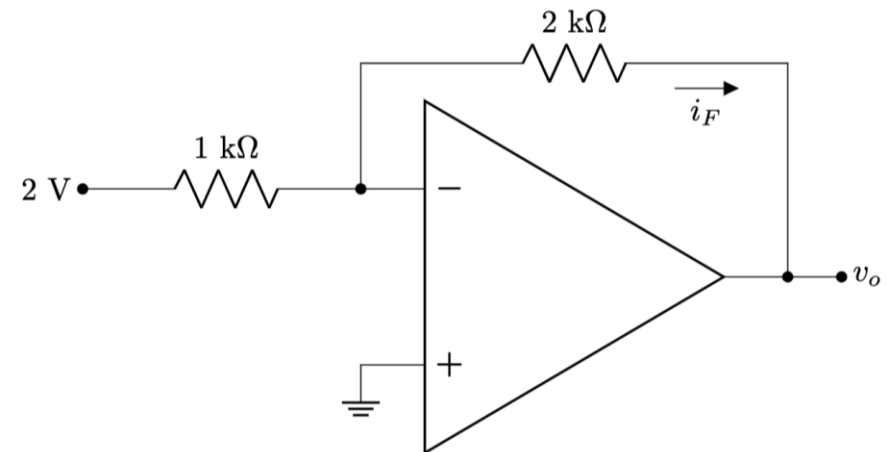
## 1. Open loop configuration:

No physical connection between input and output



## 2. Closed loop configuration:

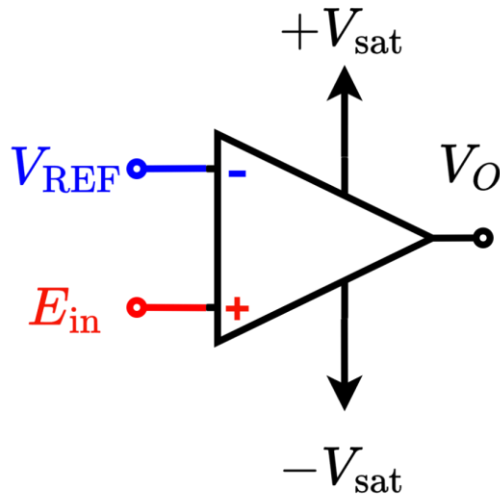
Feedback from output terminal



# Basic Op-Amp Configurations

- **Open-loop Configurations**

1. Comparator / Voltage Level Detectors

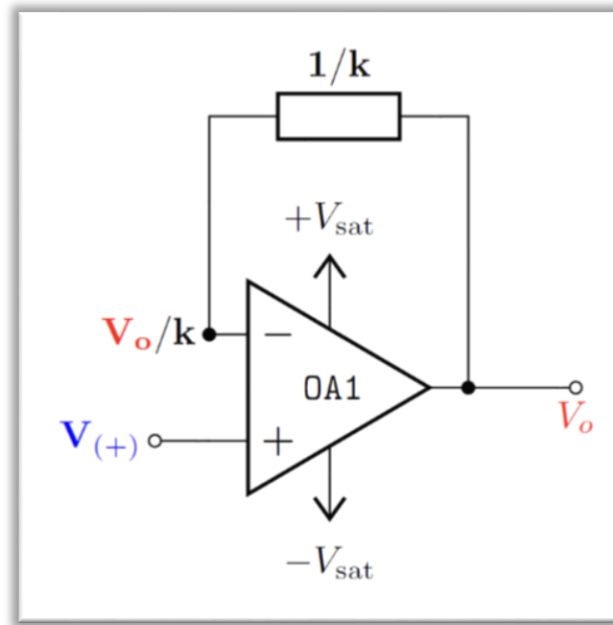


- **Closed Loop Configurations**

1. Voltage Follower
2. Inverting Amplifier
3. Inverting Summer
4. Non-Inverting Amplifier
5. Weighted Subtractor
6. Integrator
7. Differentiator
8. Exponential Converter
9. Logarithmic Converter
10. Multiplier
11. Divider

# Closed Loop Configuration

## Feedback



# Feedback in Op-Amp circuit

## Two types of feedback

### 1. Negative Feedback:

Output voltage is fed to the inputs **negatively**

The output voltage is connected to the **inverting** terminal

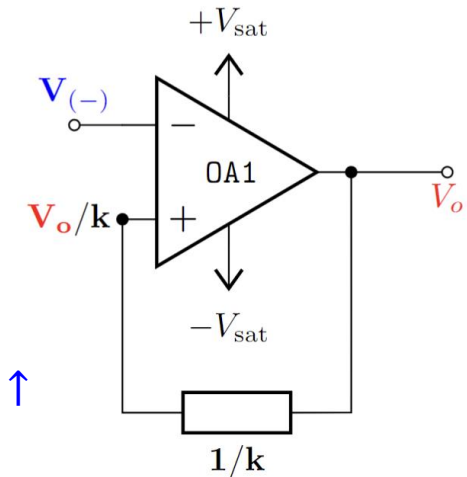
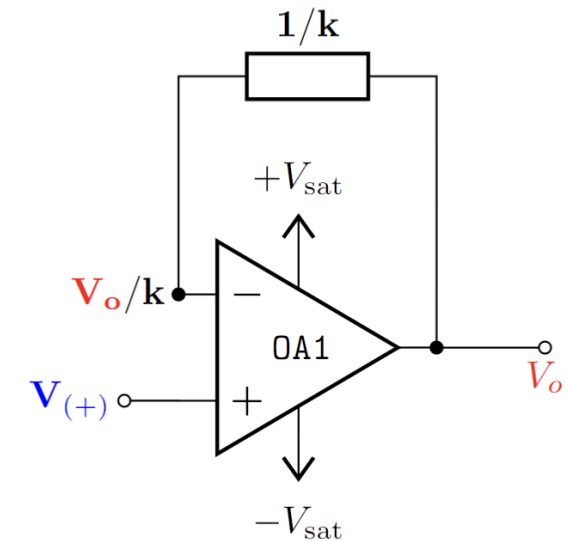
$$V_o \uparrow \Rightarrow \frac{V_o}{k} \uparrow \Rightarrow V_{(-)} \uparrow \Rightarrow V_d \downarrow = V_{(+)} - V_{(-)} \uparrow \Rightarrow V_o \propto V_d \downarrow$$

### 2. Positive Feedback:

Output voltage is fed to the inputs **positively**

The output voltage is connected to the **non-inverting** terminal

$$V_o \uparrow \Rightarrow \frac{V_o}{k} \uparrow \Rightarrow V_{(+)} \uparrow \Rightarrow V_d \uparrow = V_{(+)} \uparrow - V_{(-)} \Rightarrow V_o \propto V_d \uparrow$$



# Feedback in Op-Amp circuit

## Two types of feedback

### 1. Negative Feedback:

Output voltage is fed to the inputs **negatively**

The output voltage is connected to the **inverting** terminal

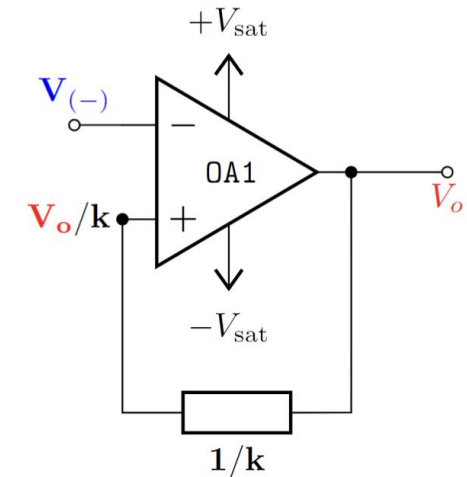
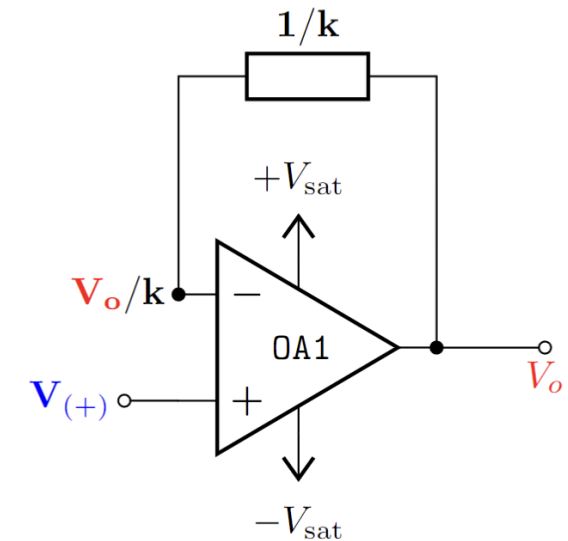
$$V_o \uparrow \Rightarrow V_o \propto V_d \downarrow$$

### 2. Positive Feedback:

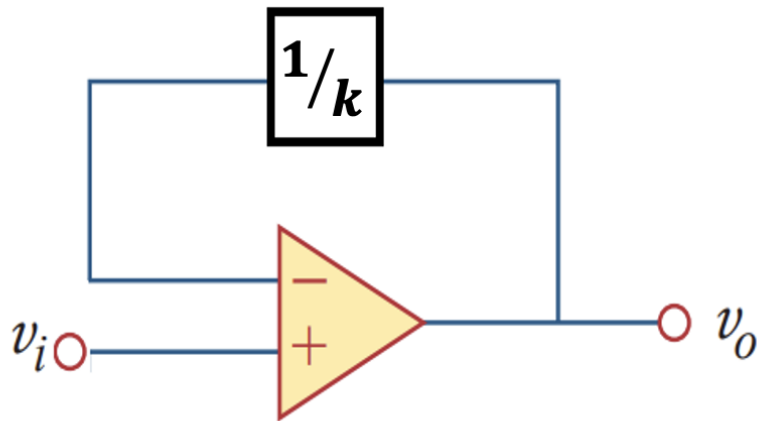
Output voltage is fed to the inputs **positively**

The output voltage is connected to the **non-inverting** terminal

$$V_o \uparrow \Rightarrow V_o \propto V_d \uparrow$$



# Negative Feedback – Derivation of Gain



Here,  $v_- = \frac{v_o}{k}$

We know,  $v_o = A v_d$

or,  $v_o = A(v_+ - v_-)$

$$= A\left(v_i - \frac{v_o}{k}\right)$$

$$= A v_i - \frac{A}{k} v_o$$

or,  $v_o\left(1 + \frac{A}{k}\right) = A v_i$

So,  $v_o = \frac{A v_i}{1 + \frac{A}{k}}$

or,  $v_o = \frac{v_i}{\frac{1}{A} + \frac{1}{k}}$

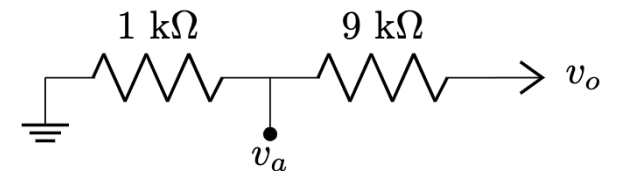
$A$  is extremely large,

so,  $\frac{1}{A} \approx 0$

$$v_o = \frac{v_i}{\frac{1}{k}} = k v_i$$

If  $k = 10$  (meaning we feed back one tenth of the output to negative input), we will get  $v_o = 10 * v_i$ . that is 10 fold gain.

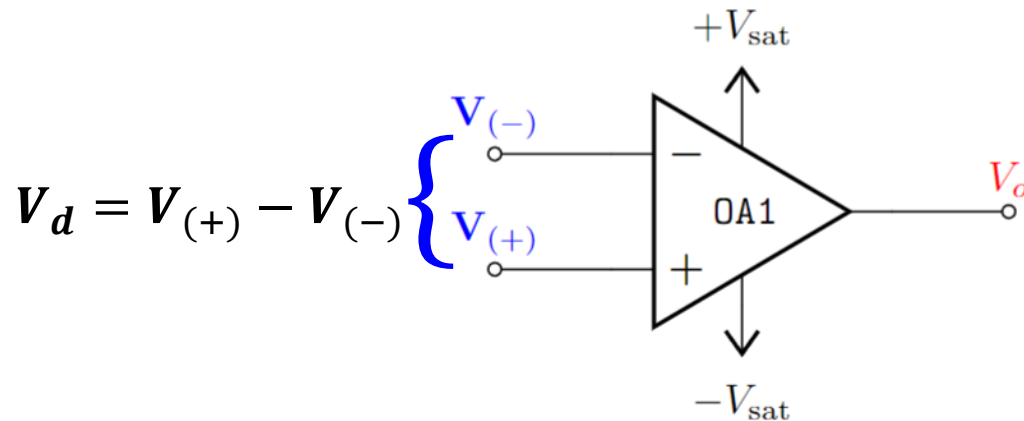
How to get  $1/k$  of output to input? Voltage dividers!



$$v_a = \frac{1 \text{ k}\Omega}{1 \text{ k}\Omega + 9 \text{ k}\Omega} \times v_o = \frac{v_o}{10}$$

# Open Loop Gain VS Closed Loop Gain

## Open Loop (OL) Configuration

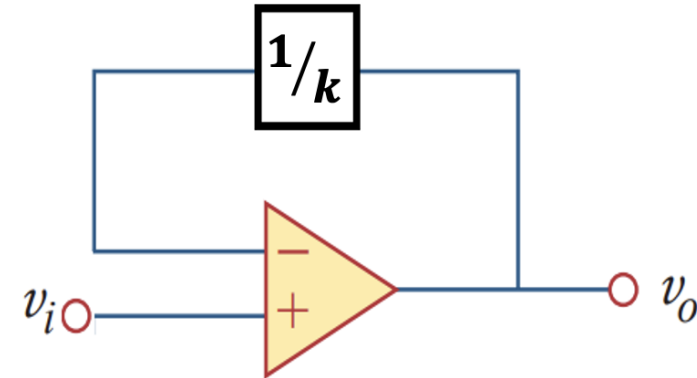


Input Voltage:  $V_d$   
Output Voltage:  $V_o$

$\therefore$  Voltage Gain:  $\frac{V_o}{V_d} = A \text{ or } K$

| OL Gain                     | CL Gain                |
|-----------------------------|------------------------|
| $A \text{ or } K \sim 10^5$ | $k \ll A$<br>$k < 100$ |

## With “Negative Feedback”: Closed Loop (CL) Configuration

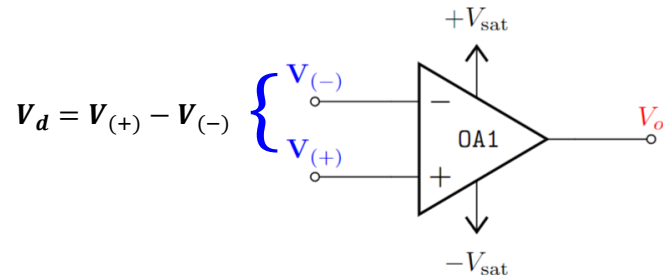


Input Voltage:  $V_i$   
Output Voltage:  $V_o$

$\therefore$  Voltage Gain:  $\frac{V_o}{V_i} = k$

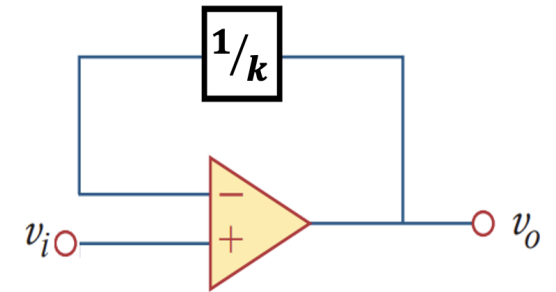
# Open Loop Gain VS Closed Loop Gain

## Open Loop (OL) Configuration



| OL Gain                                       | CL Gain                                   |
|-----------------------------------------------|-------------------------------------------|
| $\frac{V_o}{V_d} = A \text{ or } K \sim 10^5$ | $\frac{V_o}{V_i} = k \ll A$<br>$k < 100$  |
| Can't be controlled                           | Can be controlled by the feedback element |
| Used as " <b>Comparator</b> "                 | Used as " <b>Linear Amplifier</b> "       |

## Closed Loop (CL) Configuration

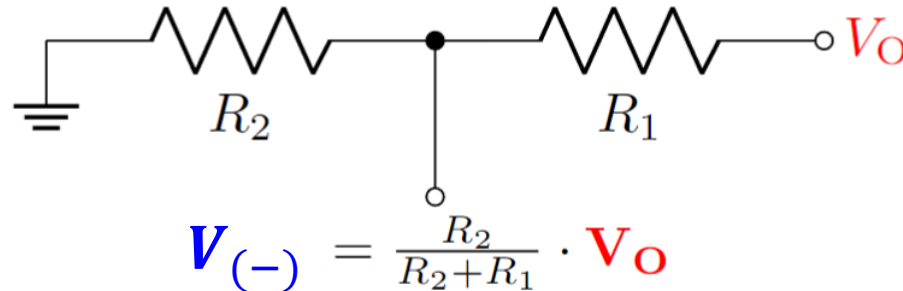


# Negative Feedback in Op-Amp circuit

The **output voltage** is transformed in the following way:

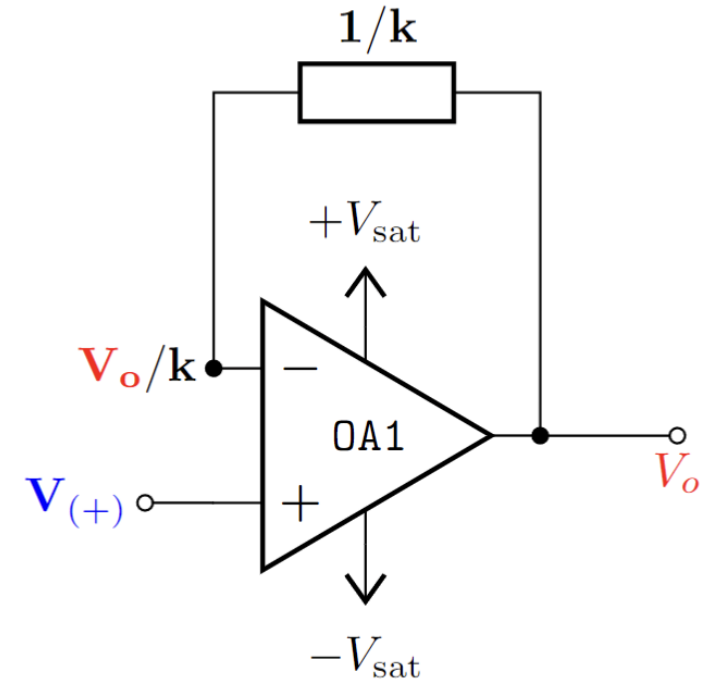
$$V_{(-)} = \frac{1}{k} \cdot V_o$$

This factor of  $1/k$  can be achieved with a voltage divider network.



$$V_{(-)} = \frac{R_2}{R_2 + R_1} \cdot V_o$$

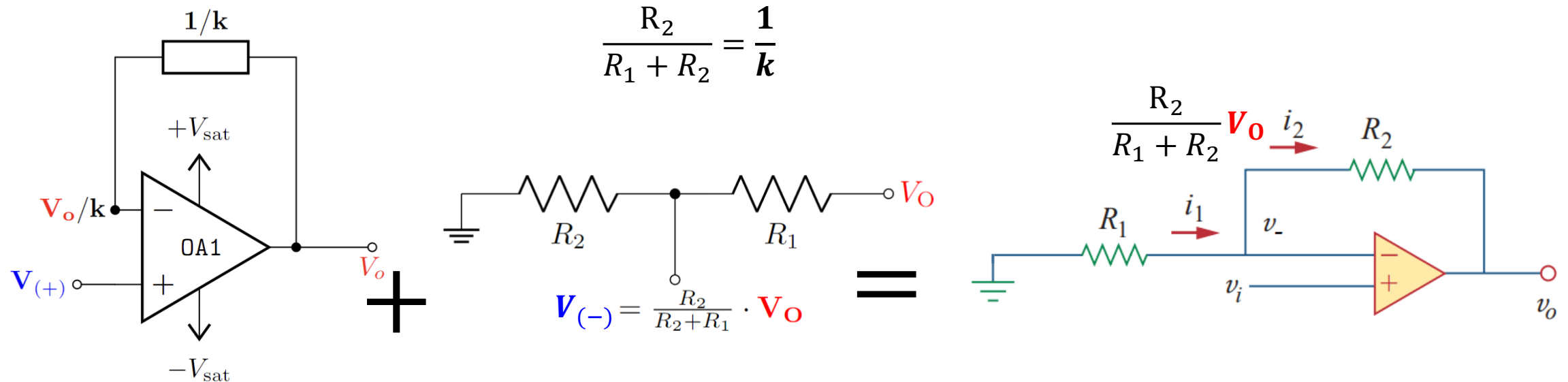
$$\frac{1}{k} = \frac{R_2}{R_1 + R_2}$$



A voltage divider can act as a multiplier/factor in the **feedback** branch

# Negative Feedback in Op-Amp circuit

A voltage divider can act as a multiplier/factor in the **feedback** branch



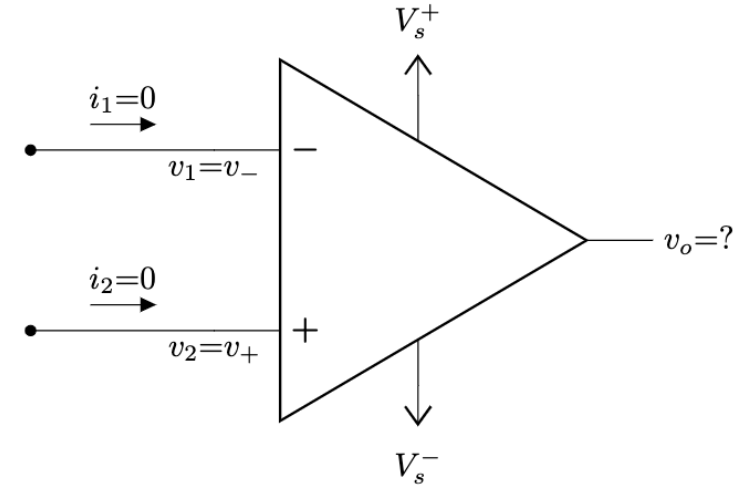
If  $k = 10$  (meaning we feed back one tenth of the output to negative input), we will get  $v_o = 10 \cdot v_i$ . that is 10-fold gain.

# Solving Closed Loop Op-Amp Circuit

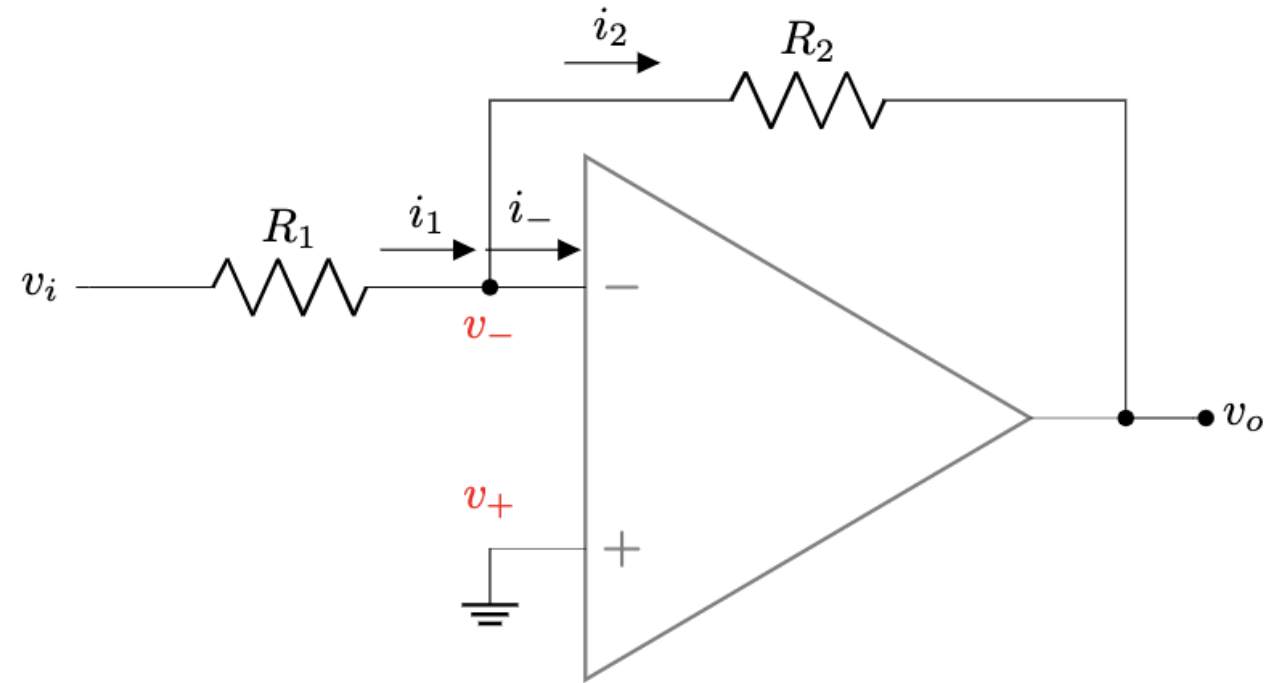
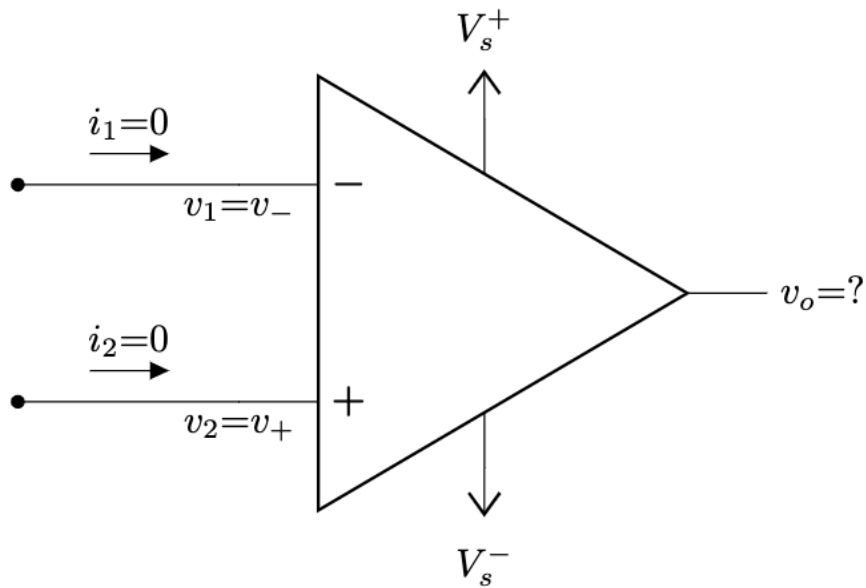
- For “**ideal**” op-amp
  - **Infinite** input resistance,  $R_i = \infty = \text{open circuit}$
  - **Zero** output resistance,  $R_o = 0 = \text{short circuit}$
  - $i_i = 0$  and  $i_+ = 0$
- **When there is negative feedback,**
  - In an ideal op-amp, “ $A$ ” (or  $K$ ) is **infinitely high**. Thus, for a finite output voltage  $v_o$ :

$$\frac{v_o}{A} = v_d \rightarrow 0 \Rightarrow v_+ = v_-.$$

- This is called **virtual short circuit**



# Solving Closed Loop Op-Amp Circuit



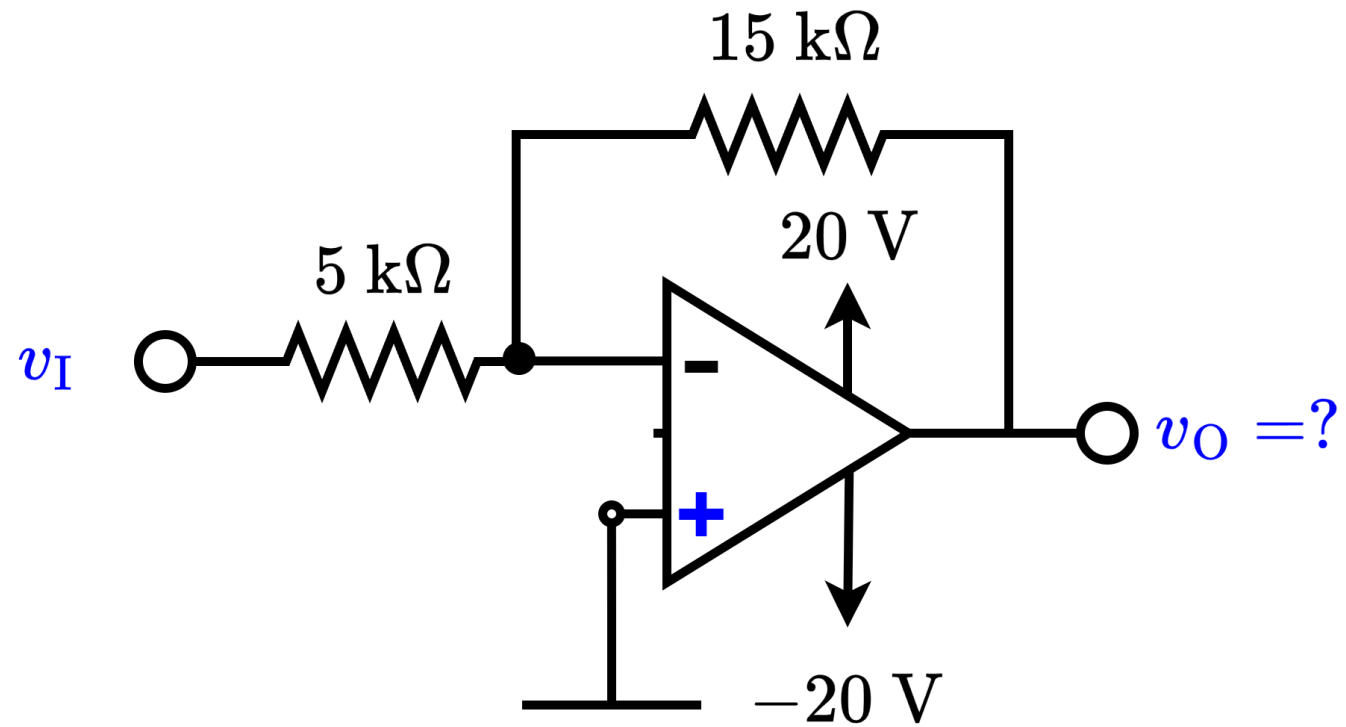
## Two Rules:

1. Virtual Shorting:
2. Zero input bias current:

$$v_+ = v_-$$

$$i_- = i_+ = 0$$

# Solving Closed Loop Op-Amp Circuit



# Solving Closed Loop Op-Amp Circuit

1. Since  $v_+$  is connected to ground,

$$v_+ = 0 \text{ V}$$

2. Since there is negative feedback, from virtual short,

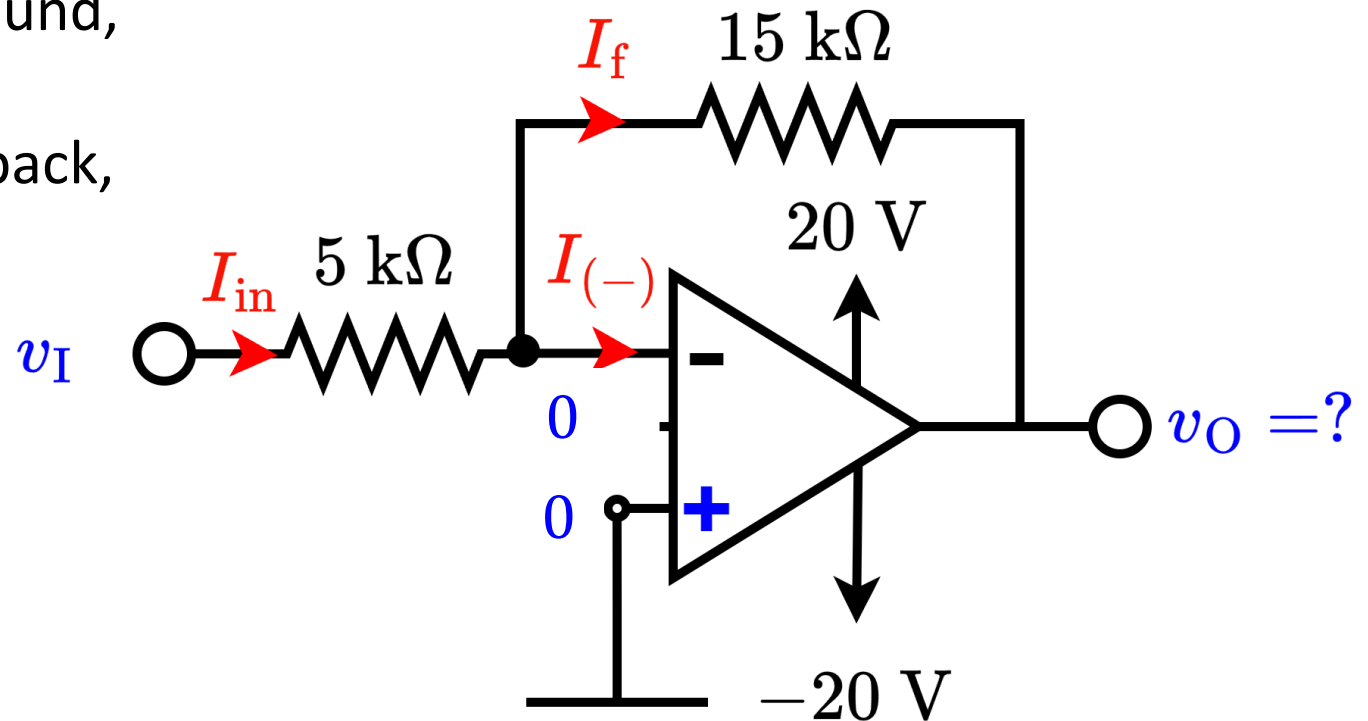
$$v_- = v_+ = 0 \text{ V}$$

3. Ohm's law for **5 k $\Omega$** :

$$I_{\text{in}} = \frac{v_I - 0}{5} = \frac{v_I}{5}$$

4. Ohm's law for **15 k $\Omega$** :

$$I_f = \frac{0 - v_O}{15} = -\frac{v_O}{15}$$



# Solving Closed Loop Op-Amp Circuit

3. Ohm's law for **5 kΩ**:

$$I_{\text{in}} = \frac{v_I - 0}{5} = \frac{v_I}{5}$$

4. Ohm's law for **15 kΩ**:

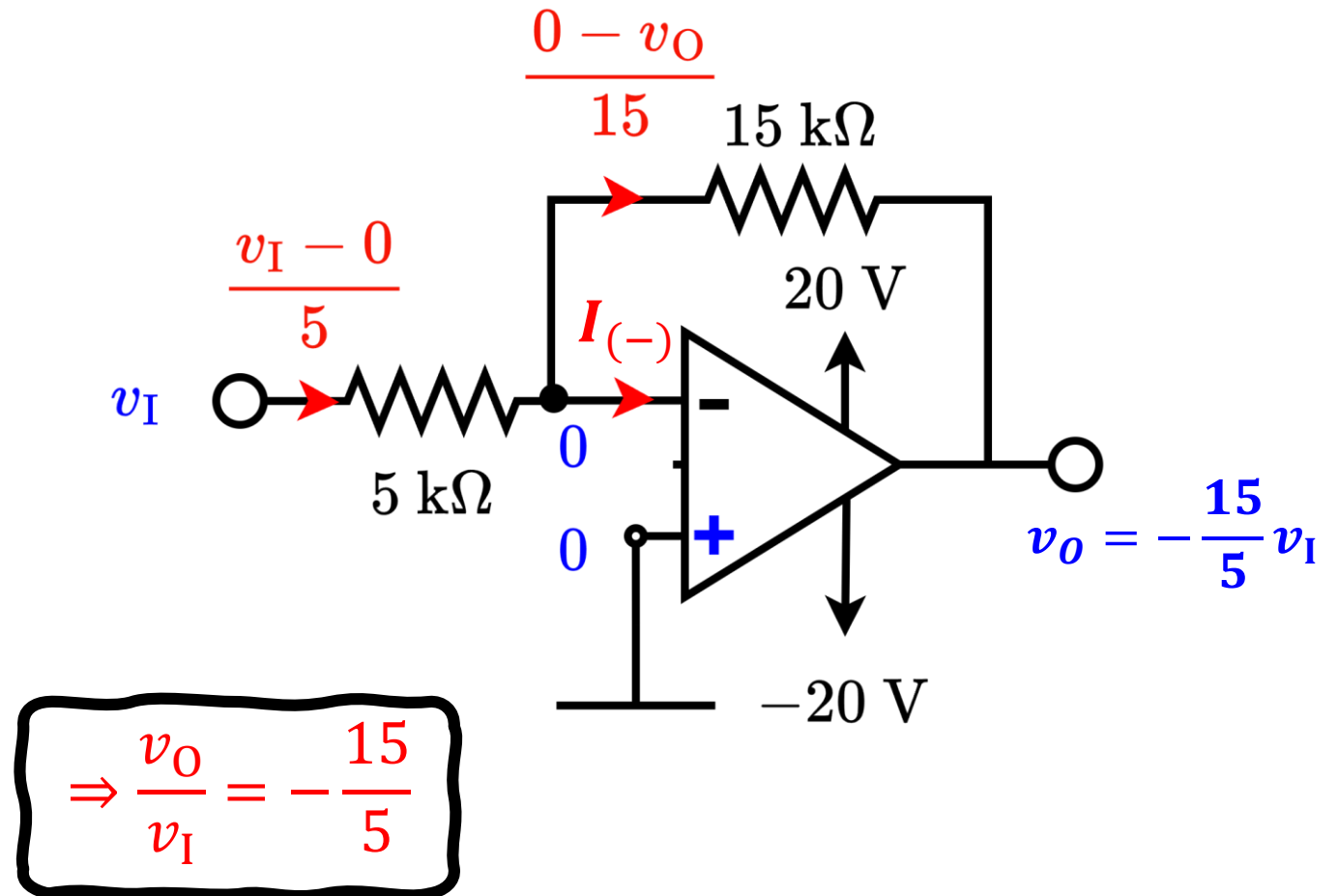
$$I_f = \frac{0 - v_O}{15} = -\frac{v_O}{15}$$

5. For “**ideal**” op-amp,

$$I_{(-)} = I_{(+)} = 0$$

6. So,  $I_{\text{in}} = I_f$

$$-\frac{v_O}{15} = \frac{v_I}{5}$$



# Solving Closed Loop Op-Amp Circuit

3. Ohm's law for **5 kΩ**:

$$I_{\text{in}} = \frac{v_I - 0}{R_I} = \frac{v_I}{R_I}$$

4. Ohm's law for **15 kΩ**:

$$I_f = \frac{0 - v_O}{R_f} = -\frac{v_O}{R_f}$$

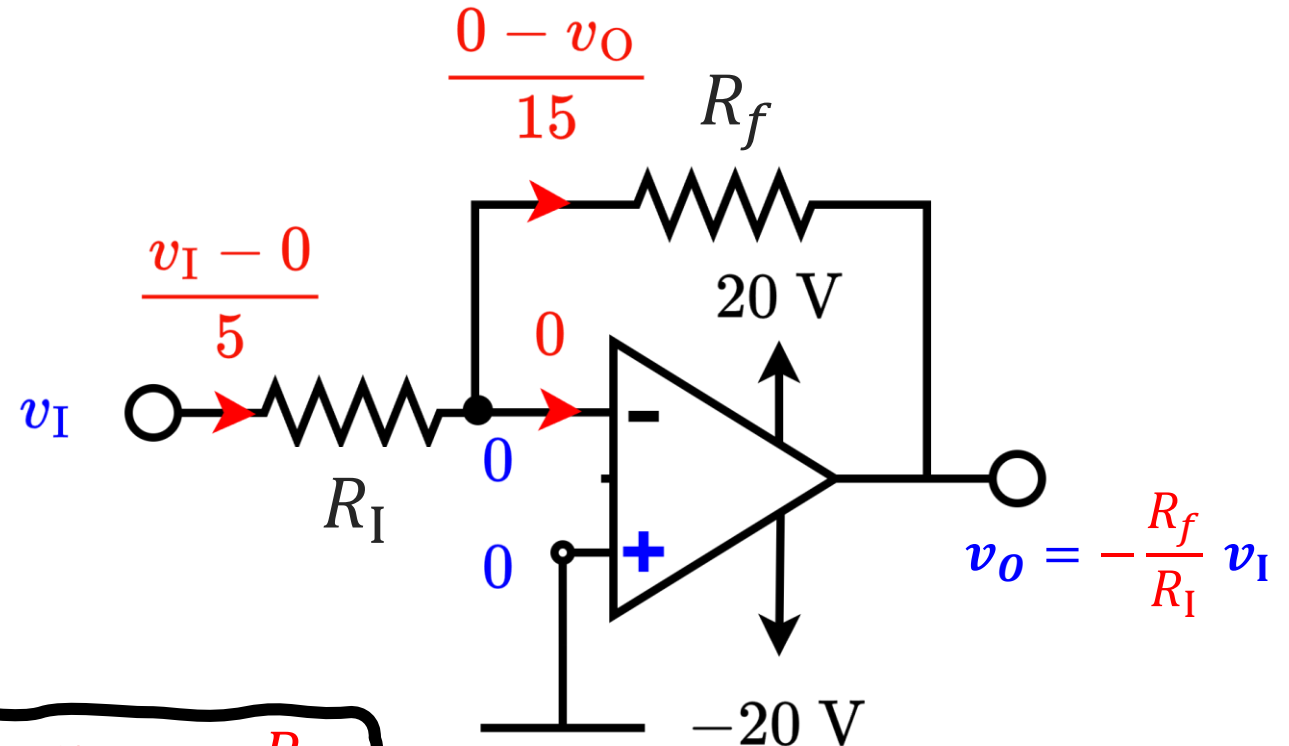
5. For “ideal” op-amp,

$$I_{(-)} = I_{(+)} = 0$$

6. So,  $I_{\text{in}} = I_f$

$$-\frac{v_O}{R_f} = \frac{v_I}{R_I}$$

$$\Rightarrow \frac{v_O}{v_I} = -\frac{R_f}{R_I}$$



# Solving Closed Loop Op-Amp Circuit

7. Check whether the amplified voltage exceeds **saturation limit**.

$$-\frac{R_f}{R_I} v_I = -\frac{15}{5} v_I = -3v_I$$

If  $v_I = 3 \text{ V}$ :

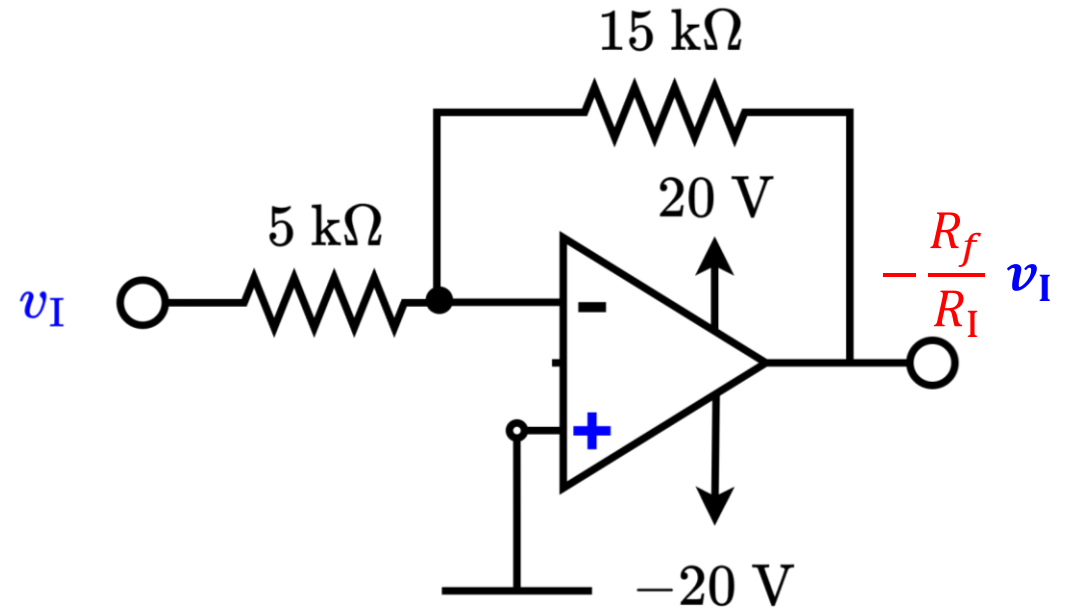
$$-3v_I = -3 \times 3 > -20 \text{ V}$$

$$\therefore v_O = -9 \text{ V}$$

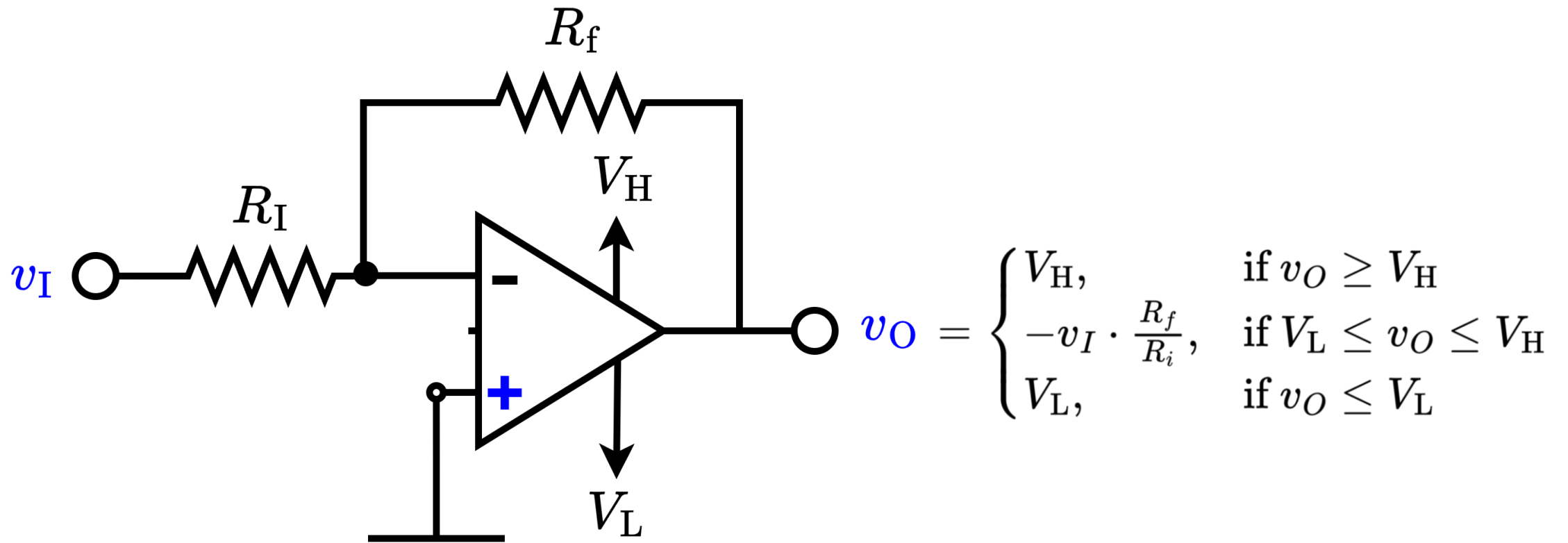
If  $|v_I| > 6.67 \text{ V}$ :

Op-amp goes into saturation as  $-3v_I = -3 \times 6.67 < -20$

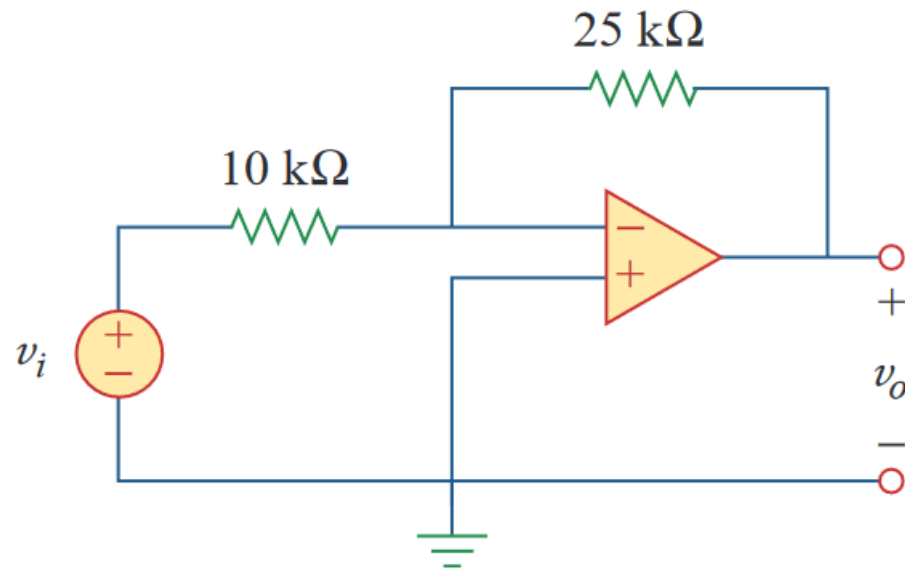
$$\therefore v_O = -20 \text{ V}$$



# Inverting Amplifier



# Example - 1



If  $v_i = 0.5\text{ V}$ , calculate:

- (a) Output voltage  $v_o$ .
- (b) Current in the  **$10\text{ k}\Omega$**  resistor.

(a)

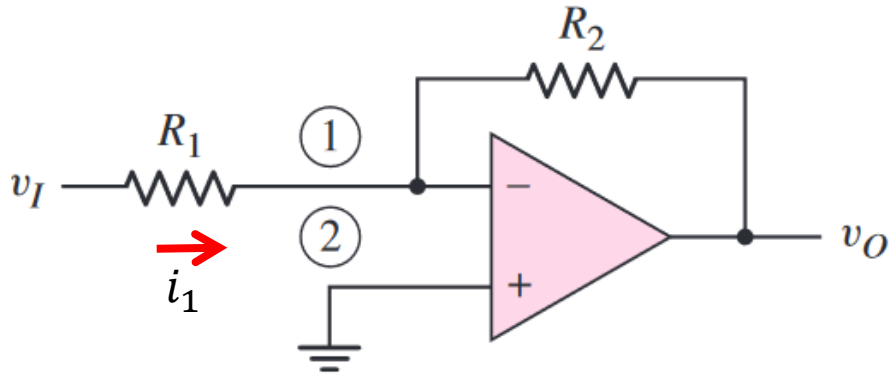
$$v_o = -\frac{R_f}{R_i} \cdot v_i = -2.5v_i = -1.25\text{ V}$$

(b) Current through the  **$10\text{ k}\Omega$**  resistor is

$$i = \frac{v_i}{R_i} = \frac{0.5}{10}\text{ mA} = 50\text{ }\mu\text{A}$$

## Example - 2

**Design** the circuit such that the closed loop voltage gain is  $A_{CL} = -5$ . Assume the op-amp is driven by an ideal sinusoidal source,  $v_I = 0.1 \sin(\omega t)$  (V), that can supply a maximum current of  $5 \mu\text{A}$ .



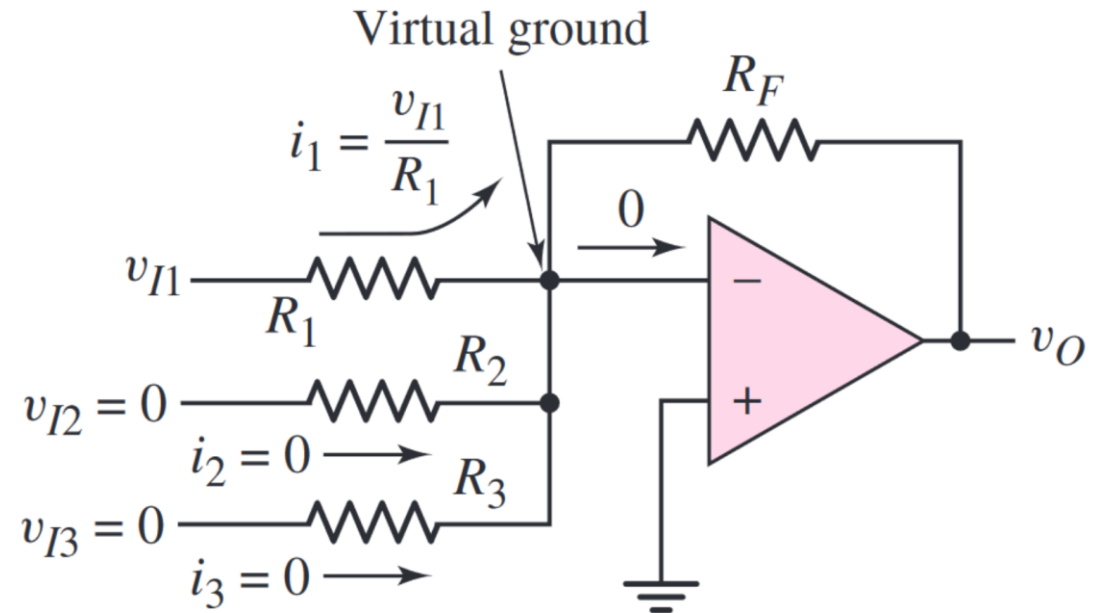
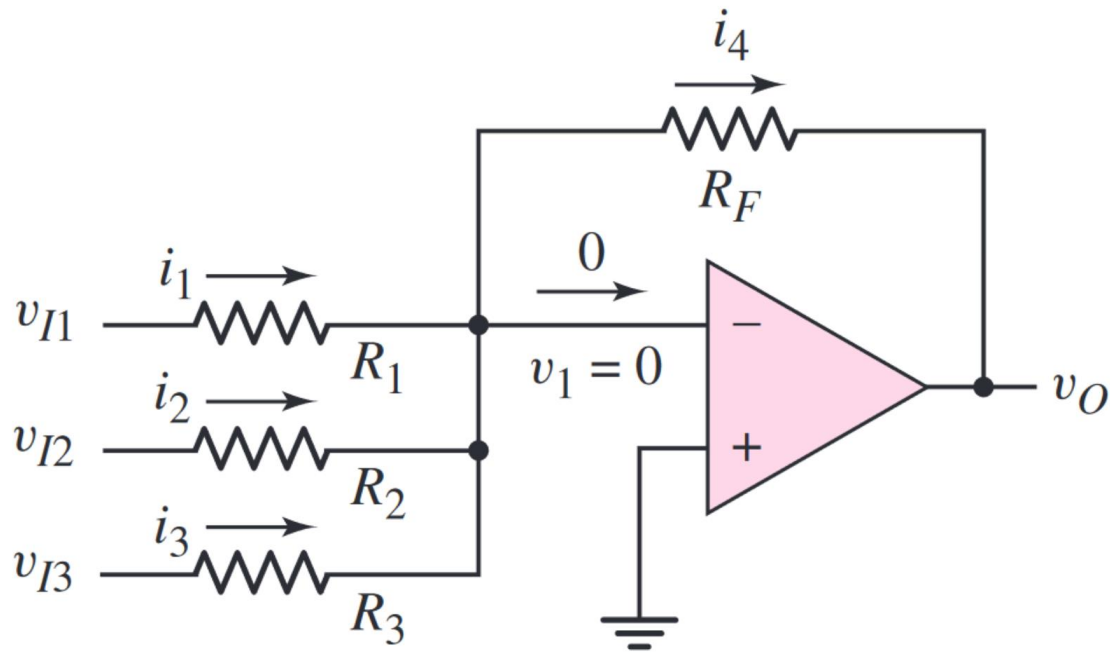
$$i_1 = \frac{v_I}{R_1}$$

$$R_1 = \frac{v_I(\max)}{i_1(\max)} = \frac{0.1}{5 \times 10^{-3}} = 20 \text{ k}\Omega$$

$$R_2 = -A_{CL} \cdot R_1 = 5 \times 20 = 100 \text{ k}\Omega$$

# Inverting Summer

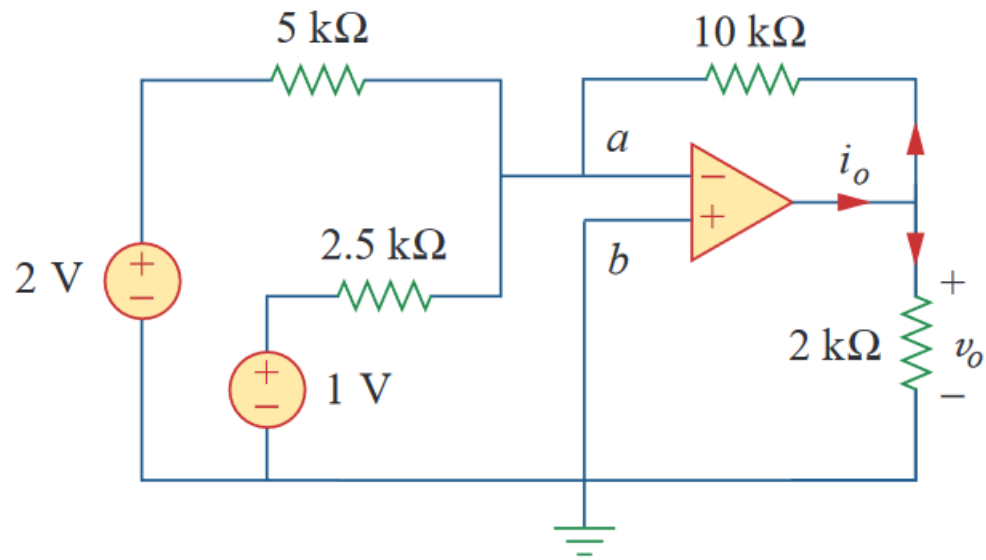
- Multichannel Amplifier



$$v_O(v_{I1}) = -i_1 R_F = -\left(\frac{R_F}{R_1}\right) v_{I1}$$

$$v_O = -\left(\frac{R_F}{R_1} v_{I1} + \frac{R_F}{R_2} v_{I2} + \frac{R_F}{R_3} v_{I3}\right)$$

# Example - 3



Calculate:

- (a) Output voltage  $v_o$ .
- (b) Output current  $i_o$ .

(a)

$$v_o = -\left(\frac{10}{5} \cdot 2 + \frac{10}{2.5} \cdot 1\right) = -8 \text{ V}$$

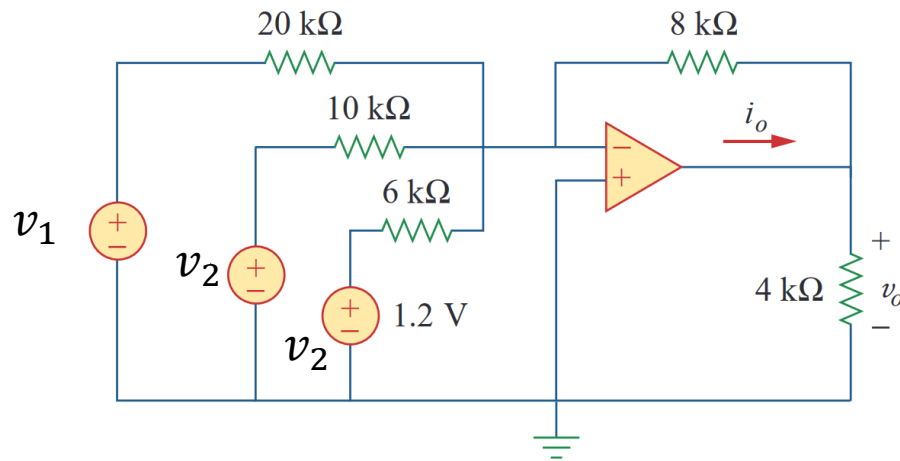
(b)

$$i = \frac{v_o}{10} + \frac{v_o}{2} = (-0.8 - 4) \text{ mA} = -4.8 \text{ mA}$$

# Example 4

Design an op-amp circuit with inputs  $v_1$ ,  $v_2$  and  $v_3$  such that, output voltage  $v_o$ :

$$v_o = -\frac{2}{5}v_1 - \frac{4}{5}v_2 - \frac{4}{3}v_3$$



## Solution:

The given function can be achieved by an **inverting summing amplifier**. Having the voltage transfer formula as:

$$v_o = \left( -\frac{R_f}{R_1}v_1 - \frac{R_f}{R_2}v_2 - \cdots - \frac{R_f}{R_n}v_n \right)$$

Here, the numerators of all the coefficients of input voltages are same ( $R_f$ ). As per the given problem, this can be achieved by setting the numerator to the LCM of 2 and 4 (i.e., to **8**).

$$\begin{aligned} v_o &= -\frac{2}{5}v_1 - \frac{4}{5}v_2 - \frac{4}{3}v_3 \\ &= -\frac{8}{20}v_1 - \frac{8}{10}v_2 - \frac{8}{6}v_3 \end{aligned}$$

# Example 4

Design an op-amp circuit with inputs  $v_1$ ,  $v_2$  and  $v_3$  such that, output voltage  $v_o$ :

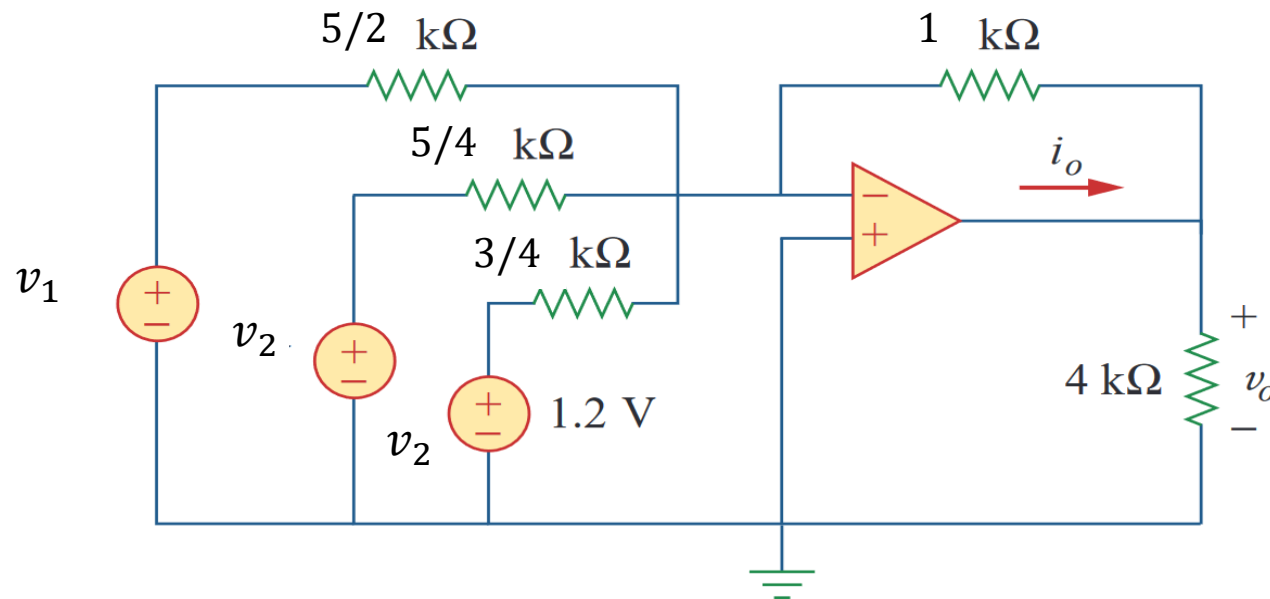
$$v_o = -\frac{2}{5}v_1 - \frac{4}{5}v_2 - \frac{4}{3}v_3$$

**Easier Solution:**

$$v_o = -\frac{2}{5}v_1 - \frac{4}{5}v_2 - \frac{4}{3}v_3$$

$$= -\frac{8}{20}v_1 - \frac{8}{10}v_2 - \frac{8}{6}v_3$$

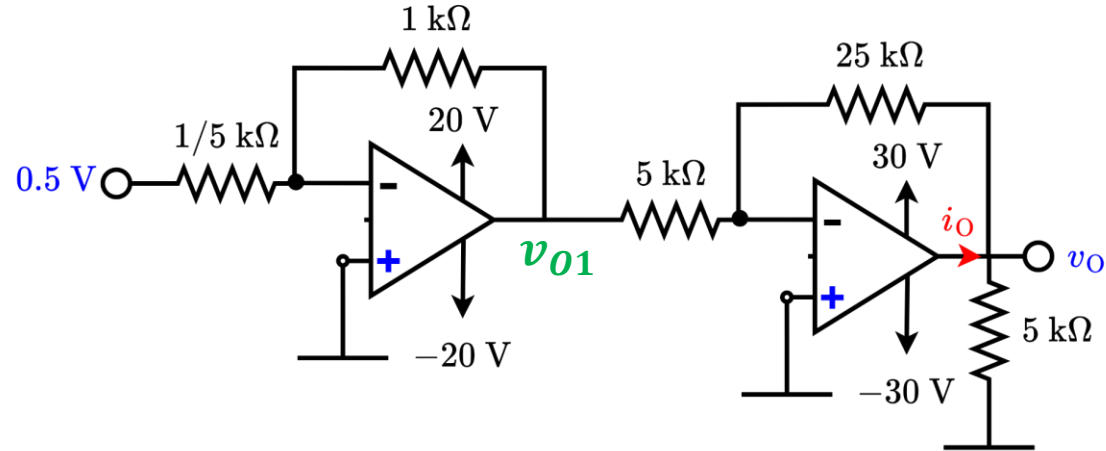
$$= -\frac{1}{5/2}v_1 - \frac{1}{5/4}v_2 - \frac{1}{3/4}v_3$$



# Example - 5

$v_I = 0.5 \text{ V}$ . Calculate:

- (a) Output voltage  $v_o$ .
- (b) Output current  $i_o$ .



(a)

$$v_{O1} = -\frac{1}{1/5} \times 0.5 \text{ V} = -2.5 \text{ V}$$

$$v_o = -\frac{25}{5} \cdot v_{O1} = 12.5 \text{ V}$$

$$v_o = \left(-\frac{1}{1/5}\right) \cdot \left(-\frac{25}{5}\right) \cdot 0.5 = 12.5 \text{ V}$$

(b)

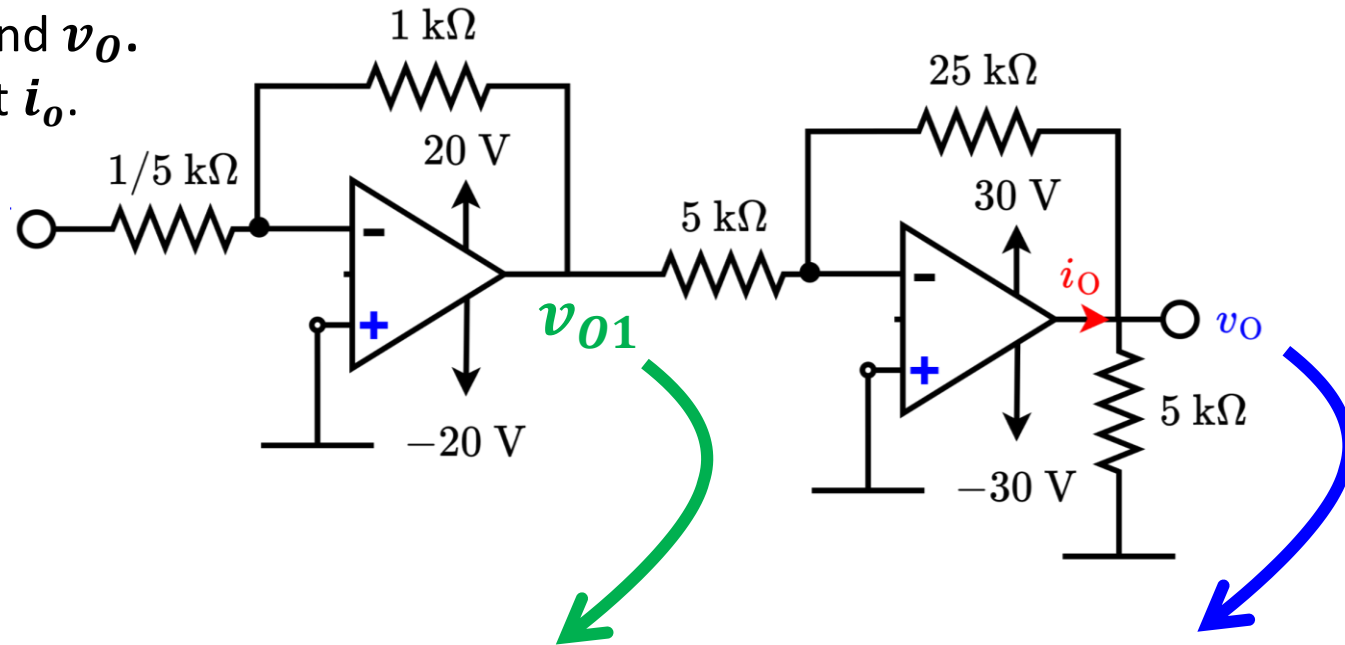
$$i = \frac{v_o}{5} + \frac{v_o}{25} = (2.5 + 0.5) \text{ mA} = 3 \text{ mA}$$

# Example - 6

Calculate:

- (a) Voltages  $v_{o1}$  and  $v_o$ .
- (b) Output current  $i_o$ .

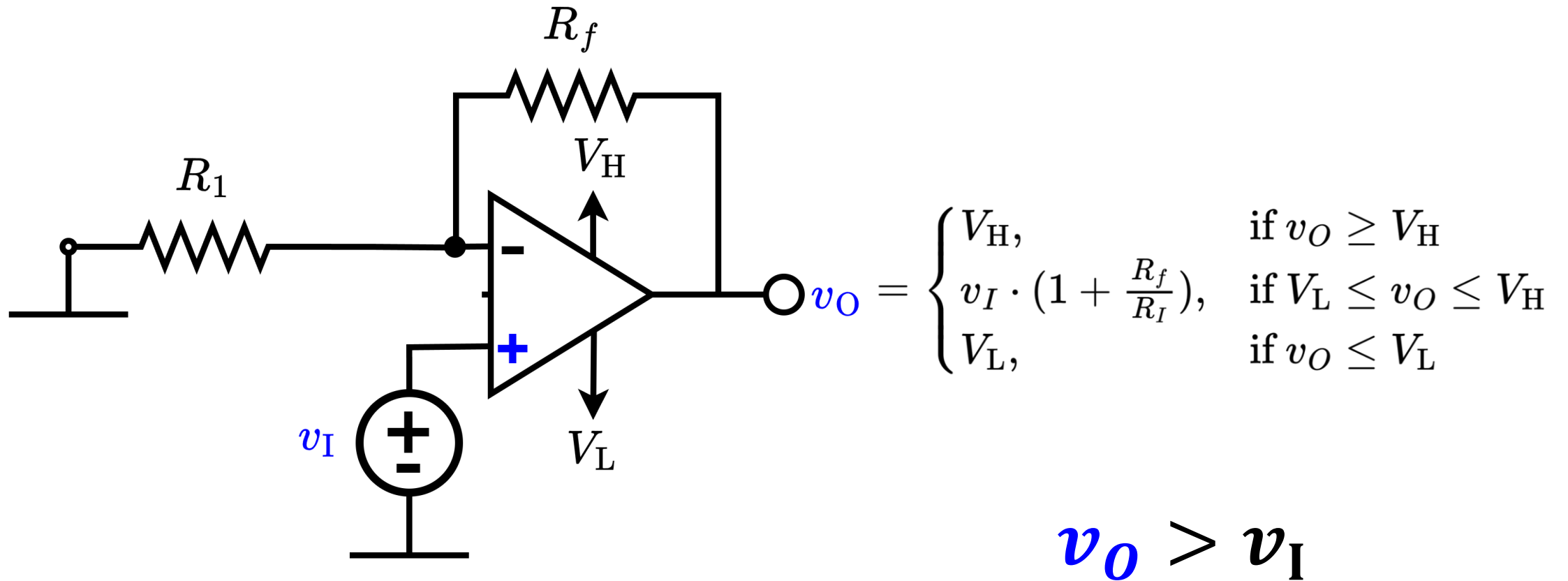
$$v_I = 0.5\sin(120\pi t)$$



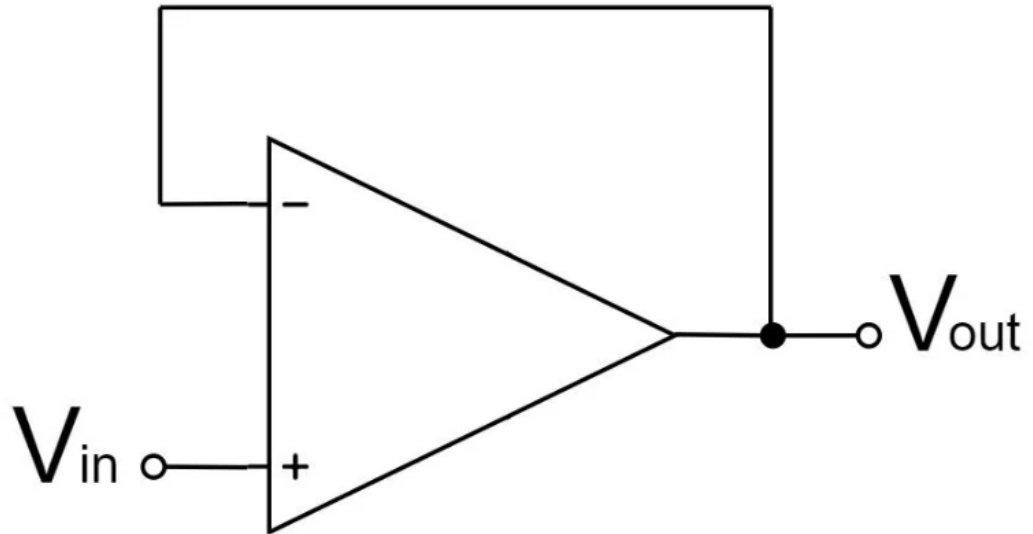
$$v_{o1} = 2.5\sin(120\pi t)$$

$$v_o = 12.5\sin(120\pi t)$$

# Non-Inverting Amplifier



# Voltage Follower / Buffer:



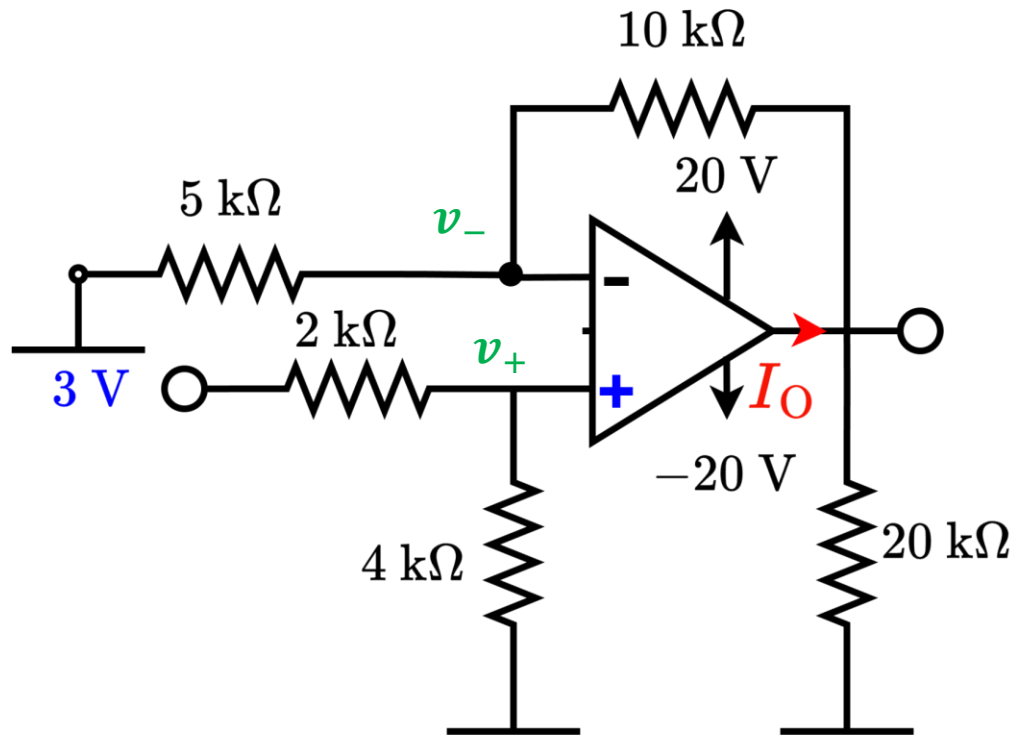
$$v_{out} = v_{in}$$

Regardless of the value of  $R_f$

# Non-Inverting Amplifier: Example 7

Calculate:

- (a) Output voltage  $v_o$ .
- (b) Output current  $I_o$ .



(a)

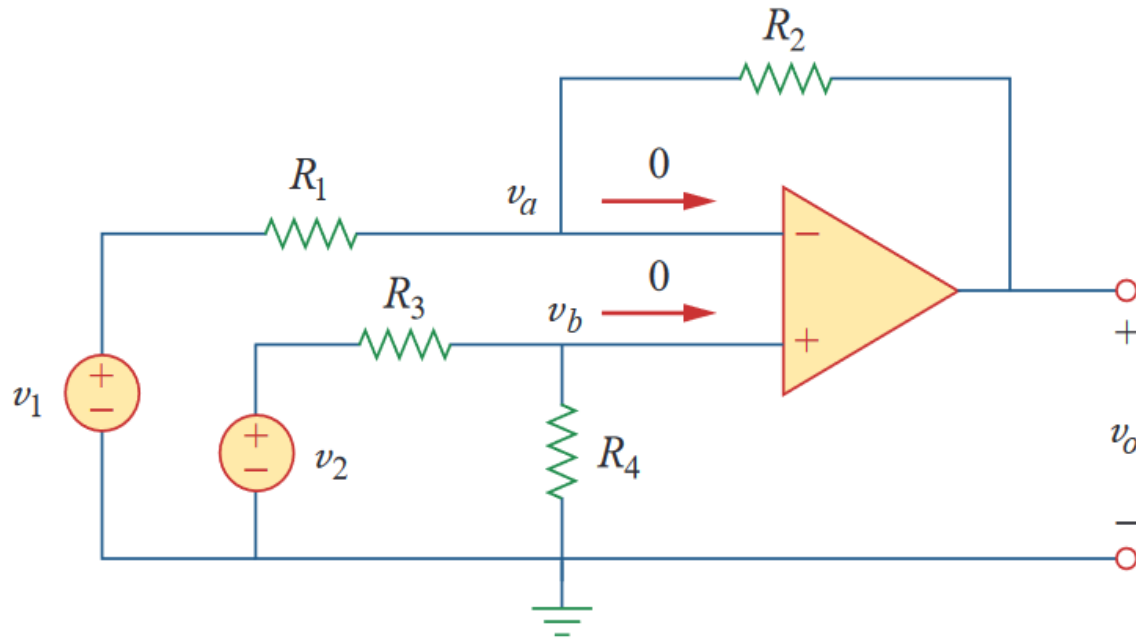
$$v_+ = \frac{4}{2 + 4} \times 3 \text{ V} = 2 \text{ V}$$

$$v_o = \left( 1 + \frac{10}{5} \right) \cdot v_+ = 6 \text{ V}$$

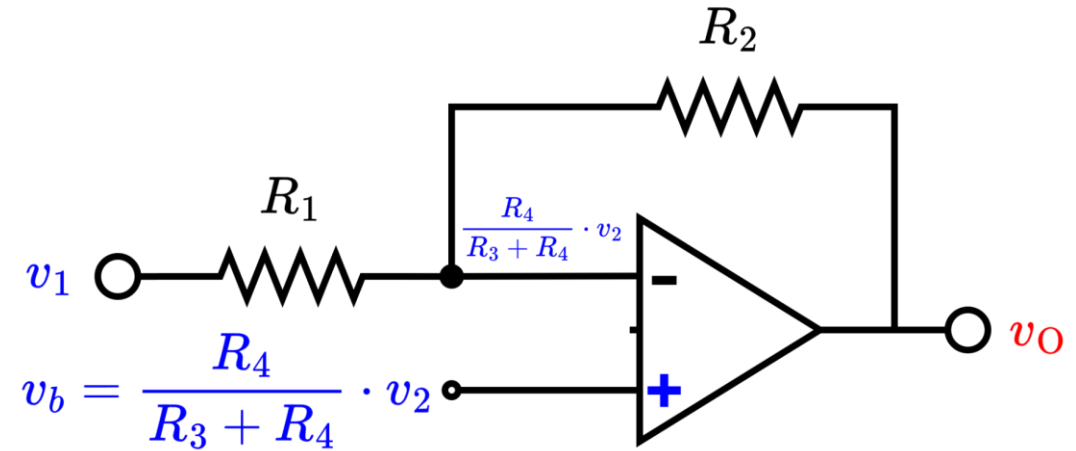
(b)

$$i = \frac{v_o}{20} + \frac{v_o}{10} = (0.3 + 0.6) \text{ mA} = 0.9 \text{ mA}$$

# Difference Amplifier



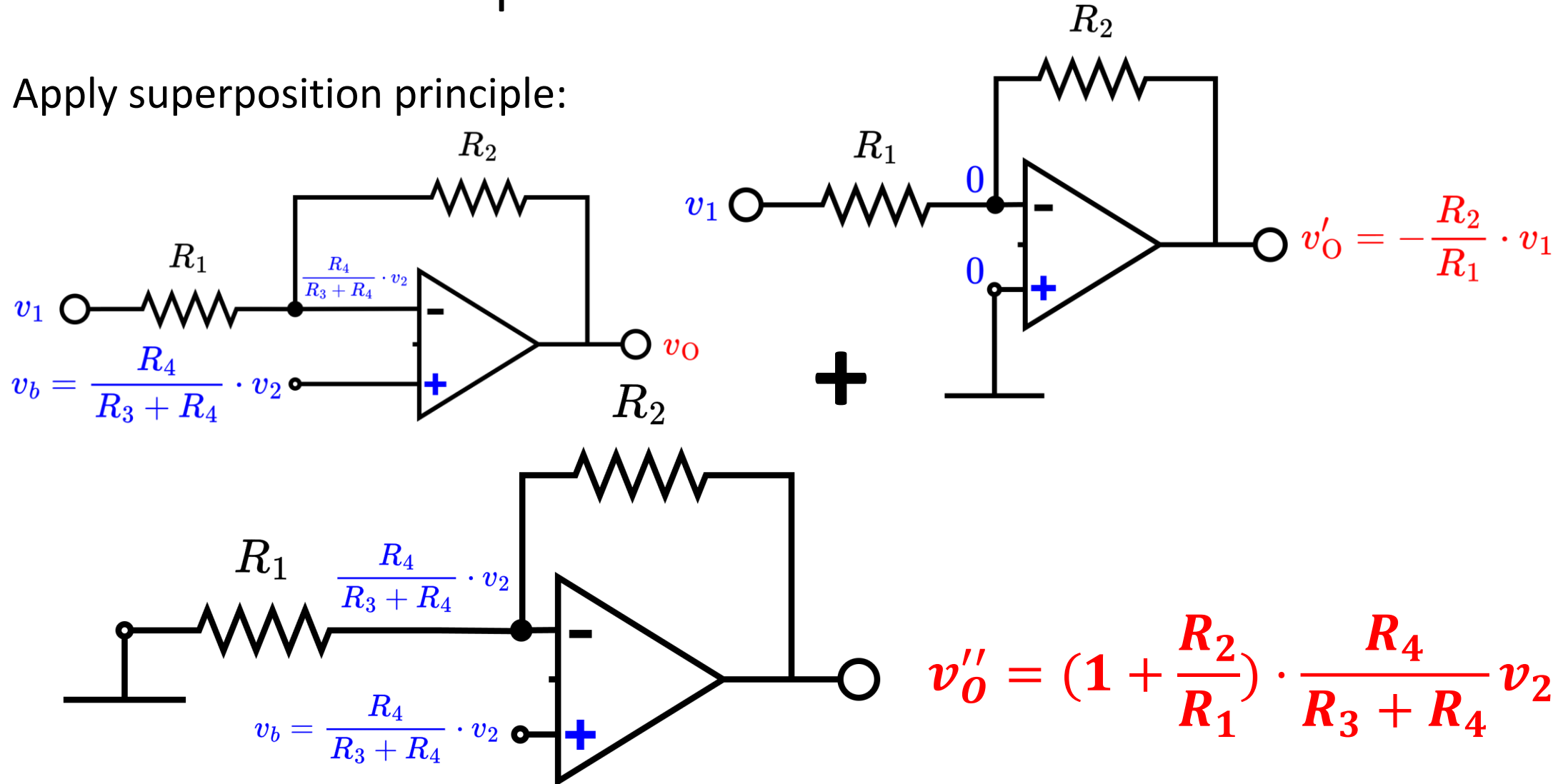
$$v_b = v_a = \frac{R_4}{R_3 + R_4} v_2$$



Simplified Circuit

# Difference Amplifier

Apply superposition principle:

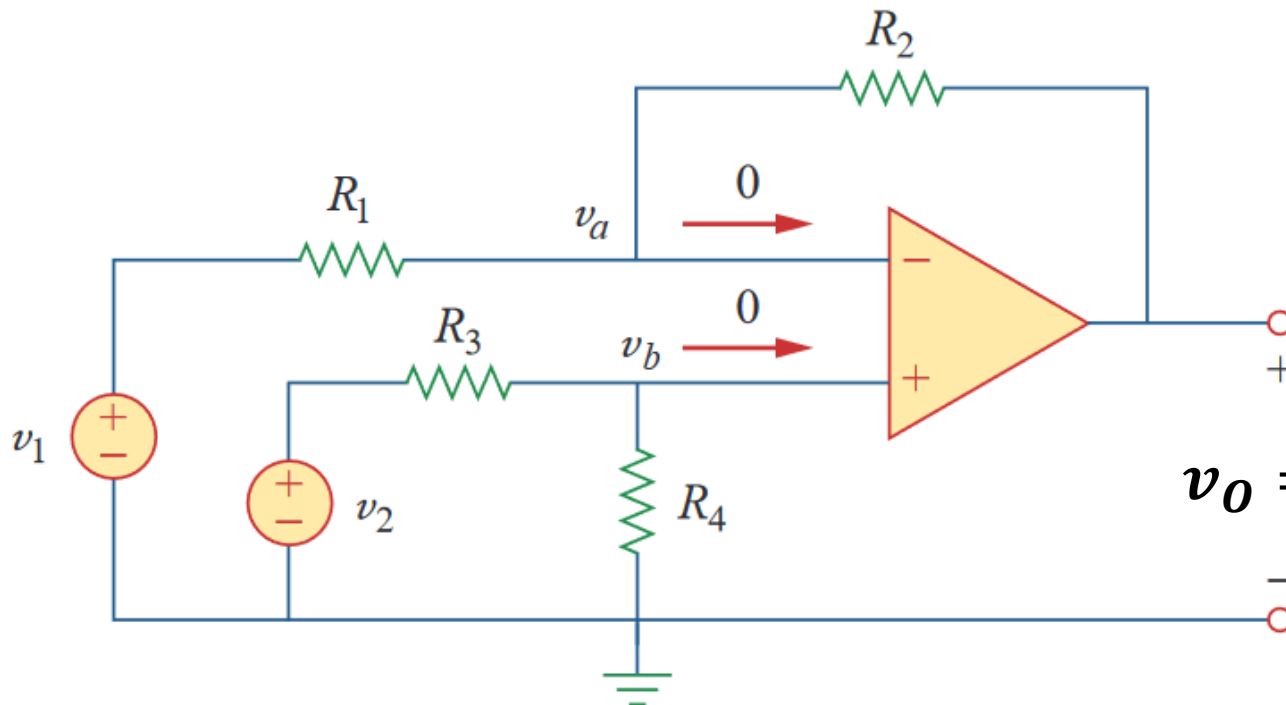


# Difference Amplifier

$$v_b = v_a = \frac{R_4}{R_3 + R_4} v_2$$

$$v_o = v_o'' + v_o'$$

$$v_o = \left(1 + \frac{R_2}{R_1}\right) v_b - \frac{R_2}{R_1} \cdot v_1$$



$$v_o = \left(1 + \frac{R_2}{R_1}\right) \cdot \frac{R_4}{R_3 + R_4} v_2 - \frac{R_2}{R_1} \cdot v_1$$

# Difference Amplifier – Example 8

Design an op amp circuit with inputs  $v_1$  and  $v_2$  such that

$$v_o = -5v_1 + 3v_2.$$

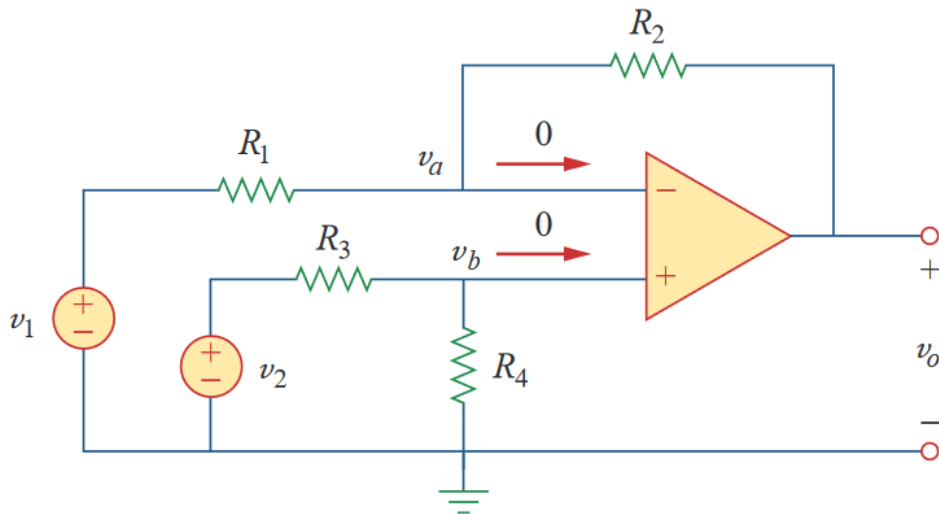
**Solution: Method 1**

$$v_o = -\frac{R_2}{R_1} \cdot v_1 + \left(1 + \frac{R_2}{R_1}\right) \cdot \frac{R_4}{R_3 + R_4} v_2$$

$$\therefore \frac{R_2}{R_1} = 5$$

$$\therefore (1 + 5) \cdot \frac{R_4}{R_3 + R_4} = 3$$

$$\Rightarrow R_3 = R_4$$

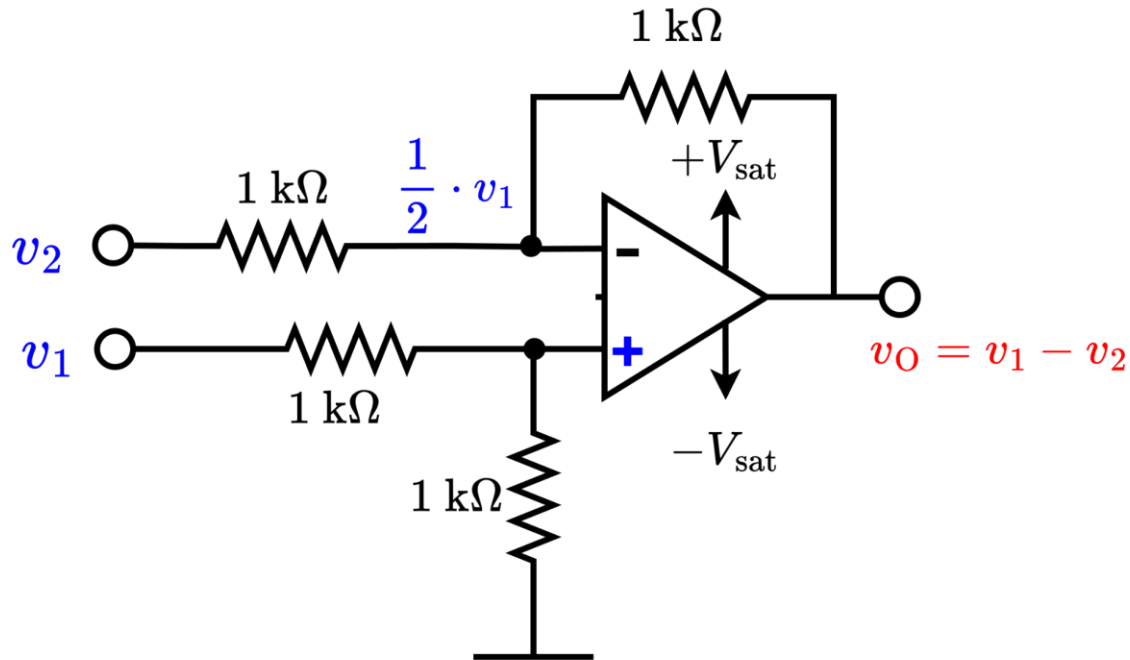


\* To implement functions of the form  $v_o = -Av_1 + Bv_2$ , with difference amplifiers, **B must be less than A + 1**

# Subtractor ( $v_1 - v_2$ )

Design an op amp circuit with inputs  $v_1$  and  $v_2$  such that

$$v_o = v_1 - v_2.$$



## Solution: Method 1

$$\text{Inverting ratio} = 1$$

$$\text{Non-inverting ratio} = 1$$

$$\text{So, } \frac{R_2}{R_1} = 1 \text{ and } \left(1 + \frac{R_2}{R_1}\right) = 2.$$

$$\therefore R_1 = R_2$$

$$\text{So, to get } \left(1 + \frac{R_2}{R_1}\right) \cdot \frac{R_4}{R_3 + R_4} = 1,$$

$$\therefore R_4 = R_3$$